The time-ordered residual plot may reveal a connection between errors occurring at one time point and errors at other time points. This connection between model errors across various time periods is termed autocorrelation.

Fig.5.3(a)shows the presence of positive autocorrelation while the presence of negative autocorrelation is demonstrated in Fig. 5.3(b)

5.4 OUTLIERS

In econometrics, outliers are data points that significantly differ from the majority of the observations in a dataset. They can have a substantial impact on statistical analyses and regression model, potentially leading to inaccurate parameter estimates and misleading conclusions. Outliers can occur for various reasons, such as data entry errors, measurement errors, or genuine extreme observations.

Let us try find out the conceptual and graphical explanation of outliers in econometrics: As you know, outliers are observations that fall far away from the typical pattern of data points in a dataset. In econometrics, outliers can manifest in different ways:

(a) **Positive Outliers:** These are observations that have values much higher than the majority of data points. Positive outliers can be caused by extraordinary events or conditions, such as a sudden spike in sales due to a promotion or a financial crisis leading to extreme stock price changes.

(b) **Negative Outliers:** Negative outliers are observations with values significantly lower than the typical data points. They might be due to errors or unusual circumstances, such as a company reporting a loss much larger than expected.

(c) **Multivariate Outliers:** In some cases, outliers might not be apparent when considering a single variable but become noticeable when you examine the relationships between multiple variables. Multivariate outliers can have a substantial impact on regression models.

In regression analysis, an outlier is defined as an observation having a notably large residual (represented as $\hat{u}_i$), which stands out when compared to the residuals of the other observations. In a simple bivariate regression, identifying such a significant residual is straightforward due to its substantial vertical deviation from the estimated regression line. It is important to note that there might be multiple outliers present. Alternatively, one can examine the value of u_i^2 to address the issue of signs (as the residuals can be either positive or negative).Among the various visualization tools available, scatter plot and box plot are the most preferred for outlier detection.

SCATTER PLOT

Fig. 5.4: A Scatter Plot showing Outlier at X=375

The scatter plot in Fig 5.4 reveals a basic linear relationship between X and Y for most of the data, and a single outlier (at X = 375).

The concern with outliers lies not only in their divergence from the bulk of the data but rather in their potential to exert a disproportionate influence on estimation techniques such as ordinary least squares. In such instances, an outlier can be categorized as an 'influential observation'.

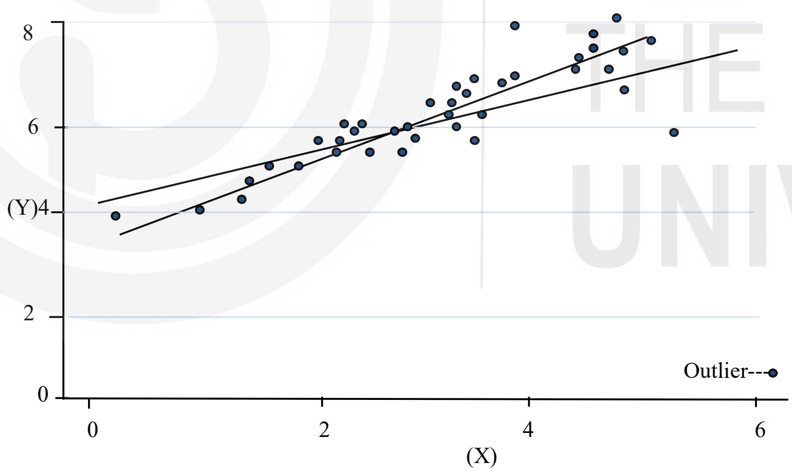

Fig. 5.5: Outlier Effects on Estimated Regression Line

One initial approach to assess the potential existence of outliers in a regression analysis involves examining the ordinary least squares (OLS) residuals, using all available observations. However, it is important to note that this method may not always provide conclusive insights. Remember that OLS relies on minimizing the sum of squared residuals as a fundamental principle, this suggests that when residuals are larger in magnitude, they incur a greater penalty.

Thus, the ordinary least squares (OLS) method aims to mitigate the presence of exceptionally large residuals. This is exemplified by the significant impact an outlier, as depicted in the figure below, can have on the estimated regression line. Consequently, it is advisable to scrutinize an observation's residual when estimating model coefficients using only the remaining portion of the dataset.

Check Your Progress 1

1) Why is it important to assess the influence of individual observations on the Ordinary Least Squares (OLS) estimation process, and what steps can be taken to address potential issues?

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2) Explain how residual plosts can help in detecting potential issues such as heteroscedasticity and autocorrelation in regression models.

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3) Why is it important to detect and address outliers in regression analysis? How do you outliers affect the techniques such a Ordinary Least Squares (OLS)?

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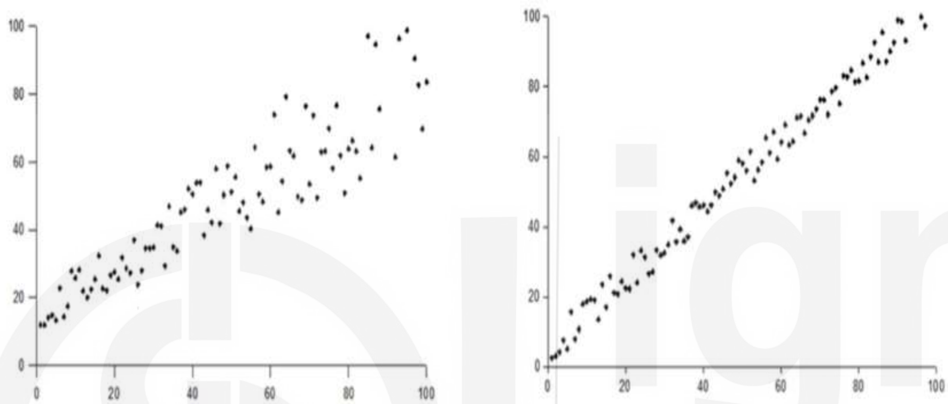
5.5 VISUAL DETECTION OF HETEROSCEDASTICITY

To assess the presence of heteroscedasticity, conduct a visual examination of residuals plotted against fitted values, or alternatively, plot the independent variable suspected to be associated with the variance of the stochastic error variable. In a large sample, an ideal scenario would reveal a consistent 'envelope' of uniform width when plotting residuals against the independent variable. In contrast, smaller samples may exhibit slightly larger residuals near the mean of the distribution compared to the extremes. Therefore, if the residuals appear to have a consistent size across all values of X, or in the case of a small sample, exhibit slightly larger residuals near the mean of X, it is generally reasonable to assume that heteroscedasticity is not a significant concern. However, if the residual plot displays an uneven envelope, where the width of the envelope significantly varies for different values of X, it is advisable to proceed with a more formal test for heteroscedasticity.

In the analysis of regression results, it is crucial to verify that the residuals exhibit consistent variance. When the residuals display different values of variance across segments we can sense the presence of heteroscedasticity (see Fig. 5.6(a)).

On the contrary, when the residuals maintain a uniform variance, we refer to this condition as homoscedasticity. Homoscedasticity indicates that the residuals possess equal variability across all independent variables [see Fig. 5.6(b)].



(a) (b)

Fig. 5.6: Heteroscedasticity

In the case of a homoscedastic model, we can reasonably assume that the residuals originate from a population with stable variance. This condition aligns with one of the fundamental assumptions of ordinary least squares (OLS) regression, contributing to a more accurate model.

5.6 VISUAL DETECTION OF AUTOCORRELATION

Autocorrelation, also known as serial correlation, occurs when the errors or residuals in a time series or regression model are correlated with their past values (lagged values). Detecting autocorrelation visually involves examining patterns in the residuals over time:

(a) **Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) Plots**: These plots are commonly used in time series analysis. ACF measures the correlation between the residuals at different lags, while PACF accounts for the correlation at each lag while removing the influence of shorter lags. Significant spikes or patterns in these plots can suggest autocorrelation.

(b) **Lag Plots:** Create a scatterplot of the residuals against their lagged values (residuals at previous time points in time series data). If you see points clustering together or forming a pattern, it may indicate autocorrelation.

(c) **Time Series Plots:** For time series data, simply plotting the residuals over time and visually inspecting for any trends or repeating patterns can provide insights into autocorrelation. If the Time Sequence Plot displays no discernible pattern, it suggests the absence of autocorrelation (as shown in Fig. 5.7(e)). However, if the plot reveals a pattern, it indicates the presence of autocorrelation, as illustrated in Fig. 5.7.

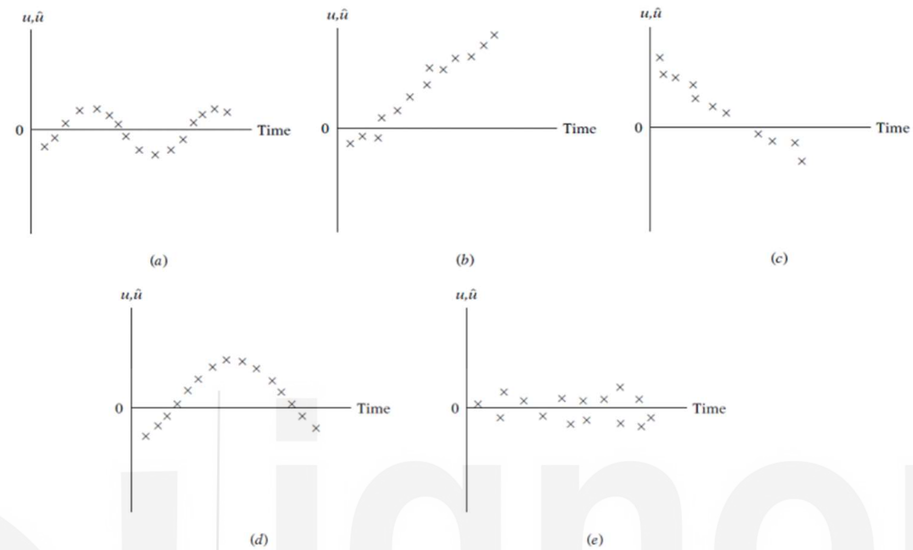


Fig. 5.7: Presence of Autocorrelation

Visual detection of heteroscedasticity and autocorrelation is a preliminary step in model diagnostics. If you suspect the presence of these issues based on visual examination, it is important to follow up with formal statistical tests to confirm and quantify the extent of these problems in your data or model residuals.

5.7 TEST FOR NORMALITY

In the classical normal linear regression model (CNLRM), which is an extension of the Classical Linear Regression Model (CLRM), one of the critical assumptions is that the error term, represented as u_i in the regression model, follows a normal distribution. This assumption holds particular significance when dealing with relatively small sample sizes, as widely used significance tests such as t-test and F-test rely on the normality assumption. Therefore, it becomes essential to assess whether the error term conforms to a normal distribution.

The easiest way to check whether the errors are normally distributed is to use the P-P (Probability-Probability) plot. A P-P plot assesses how closely a theoretical distribution models a given data distribution. A normal probability plot of the residuals is a scatter plot with the theoretical percentiles of the normal distribution on the x-axis and the sample percentiles of the residuals on the y-axis. The comparison line is the 45 degree diagonal line. The diagonal line which passes through the lower and upper percentiles of the theoretical distribution provides a visual aid to help assess whether the relationship between the theoretical and sample percentiles is linear. The distributions are equal if and only if the plot falls on this line. Any deviation from this line indicates a difference between the two distributions.

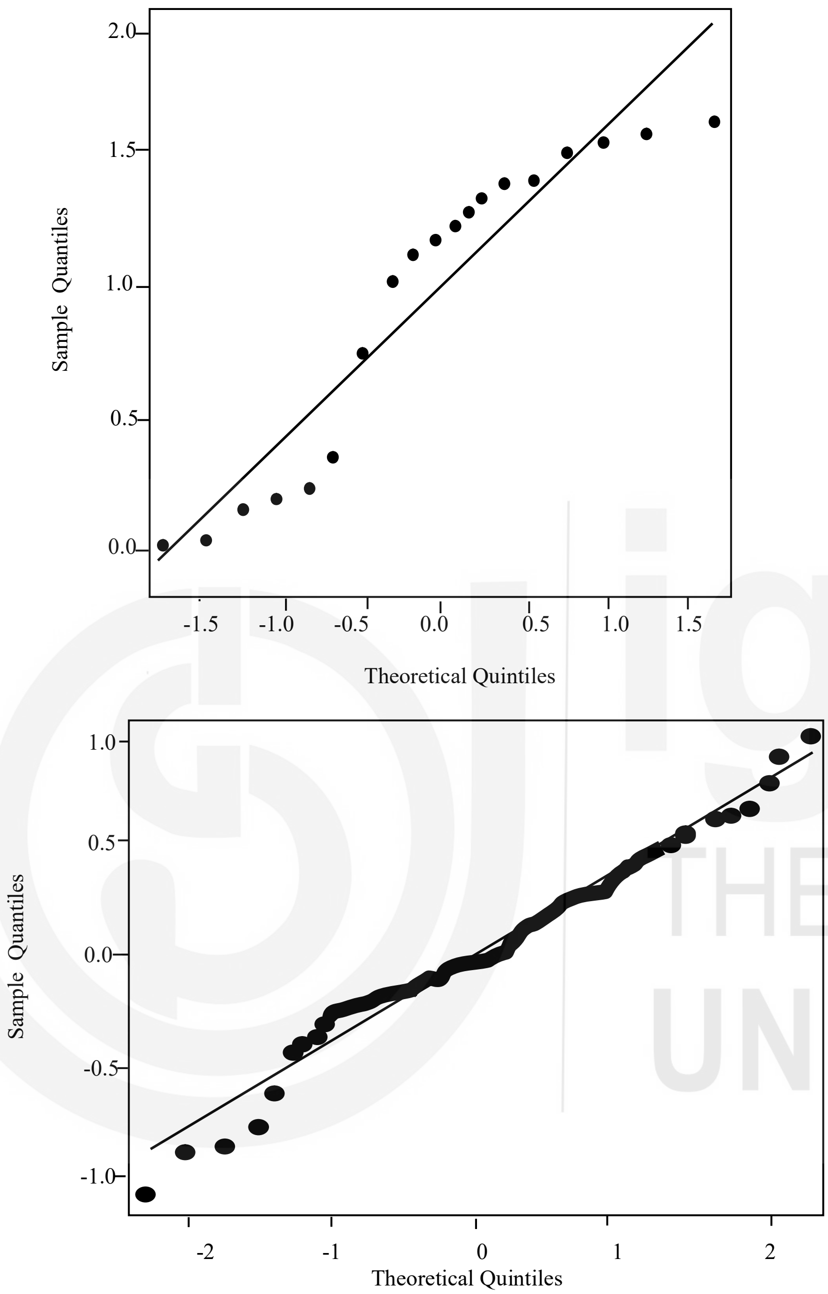


Fig. 5.8: Assessing Normality Using Probability Plot

Fig. 5.8(a) is an example of the P-P plot where the error distribution is not normal while Fig. 5.8(b) is the P-P plot where the error distribution is approximately normal.

Various formal tests of normality are available, with the Jarque–Bera (JB) test being one of the most commonly employed. However, it is important to note that the JB test is designed for large samples and may not be suitable for smaller sample sizes.

The test statistic JB is calculated using the sample skewness and kurtosis. The formula for the Jarque-Bera test statistic is as follows:

$$JB = \frac{n}{6}\left(\frac{S^2}{6} + \frac{(K-3)^2}{24}\right) \qquad \dots (5.2)$$

where:

n is the sample size.

S is the sample skewness.

K is the sample kurtosis.

In equation (5.2), if the data follows a normal distribution, JB should be approximately equal to zero. Departures from zero indicate a departure from normality. JB follows a chi-squared distribution with *two* degrees of freedom (df = 2) under the null hypothesis of normality. Therefore, you can compare the calculated JB statistic to the chi-squared distribution to determine whether the data significantly deviates from normality.

Typically, if JB statistic exceeds a critical value from the chi-squared distribution at a specified significance level (e.g., 0.05 or 0.01), it suggests that the data significantly departs from a normal distribution, indicating non-normality.

5.8 CERTAIN SPECIAL CASES

In this section we discuss certain issues on regression models, as you can see from the sub-section headings.

5.8.1 Regression through the Origin

While the majority of regression models typically incorporate a constant or intercept term, occasionally, economic theory or logical reasoning may imply that the intercept term (b0) should equal zero. In such cases, it is worth discussing Ordinary Least Squares (OLS) estimation when the intercept is constrained to be zero. In practical terms, this means that you are modelling the relationship between variables without allowing for a constant offset in the form of an intercept. For instance, the intercept-free linear model

$$Y_i = \beta X_i + u_i \qquad \dots (5.3)$$

is equivalent to the model

$$Y_i = \alpha + \beta X_i + u_i$$

when the intercept 'α' is taken to be zero.

An important characteristic of the zero-intercept model is that it employs raw sums of squares and cross-product terms. In this context, "raw" means that these terms are not adjusted by subtracting the mean values of the variables. This stands in stark contrast to the OLS regression model with an intercept, where the sums of squares and cross-product terms are mean-adjusted, meaning they are represented as deviations from the mean values of the variables.

Unless strong theoretical justifications exist, it is generally not advisable to omit the intercept term in a regression model, for several compelling reasons. Firstly, in a regression model with an intercept term, the sum of residuals always equals zero, whereas this is not guaranteed in a zero-intercept regression model. Secondly, the R-squared values for models with and without intercepts are not directly comparable. In the model with an intercept, R-squared represents the proportion of variation explained by the regression relative to the total variation around the mean of Y. In contrast, for the no-intercept model, R-squared signifies the proportion of variation explained relative to the variation around the origin (i.e., zero).

5.8.2 Change in Origin and Scale of the Variables

Typically, it is straightforward to discern how the intercept and slope estimates change when the units of measurement for the dependent variable are altered. If the dependent variable undergoes multiplication by a constant 'c' – meaning each sample value is multiplied by 'c' – then the OLS intercept and slope estimates likewise multiply by 'c'. This assumption holds when no alterations occur with the independent variable. In general, altering the units of measurement for only the independent variable does not have any bearing on the intercept. As discussed in the previous section, we defined R-squared as a goodness-of-fit metric for OLS regression. Also note that the model's goodness-of-fit remains unaffected regardless of changes in variable units.

Regression coefficients are independent of change of origin but not of scale. By origin, we mean that there will be no effect on the regression coefficients if any constant is subtracted from the value of X and Y. By scale, we mean that if the value of X and Y is either multiplied or divided by some constant, then the regression coefficients will also change.

The sensitivity of regression results to the unit of measurement varies; certain outcomes are affected, while others remain unaffected. Certain important points to note include- First, R-squared value remains constant across these regressions as R-squared is a dimensionless measure, independent of the units in which both the dependent variable (Y) and the independent variable (X) are gauged. Second, the intercept term consistently aligns with the units of measurement of the dependent variable. It is crucial to recall that the intercept signifies the dependent variable's value when the independent variable is zero. Third, when both Y and X are measured using identical units, the slope coefficients, along with their standard errors, stay consistent. However, there are variations in the intercept values and their corresponding standard errors. Fourth, in instances where the Y and X variables are expressed in distinct units, the slope coefficients exhibit dissimilarity, yet the interpretation remains unaltered.

Therefore, the units of measurement for the dependent variable (Y) and the explanatory variables (the X's) influence the interpretation of the regression coefficients. To circumvent this issue, we can represent all variables in a standardized form, where standardization is achieved by subtracting the mean value of each variable from its individual values and then dividing the difference